

PROJECT TOPIC: PHENOLOGICAL STUDY OF SURROUNDING TREES

Topic Description: Longitudinal Phenological Study of Ten Tree Species: Tracking Leaf Emergence, Flowering, Fruiting, Leaf Fall, and Avian Associations Over Seasons

Phenology is the scientific study of periodic, recurring biological lifecycle events in plants and animals and their precise climatic drivers, such as seasonal changes in temperature, day-length, and rainfall. In botany, tree phenology tracks the rhythmic timing of leaf emergence (bud burst), foliage maturation, flowering, fruiting, seed dispersal, and winter leaf senescence. These rhythmic cycles are finely tuned biological clocks developed over millions of years of evolution.

This project documents a systematic observational study of **Tree Phenology**, monitoring **Ten Distinct Tree Species** (incorporating both indigenous keystone trees and widely planted exotic ornamentals) situated in our immediate neighborhood over consecutive seasonal transitions in Maharashtra.

In the contemporary context of anthropogenic climate change, tree phenology has emerged as one of the most sensitive bio-indicators of global warming. By tracking seasonal changes in canopy cover and recording visiting insect pollinators and nesting avifauna, this project examines how changing local weather patterns influence floral reproductive cycles and urban wildlife survival.

2. Importance of the topic (विषयाचे महत्त्व)

Phenological documentation of trees provides indispensable scientific insights into climate change and ecosystem services:

Scientific & Ecological Significance :-

- **Sensitive Biological Indicator of Global Climate Warming:** Tracking the exact week of flowering over multiple years reveals phenological shifts. Warming spring temperatures cause trees to flower 10 to 15 days earlier than historic averages, demonstrating localized climatic warming.
- **Preventing Trophic Ecological Mismatches:** Many migratory birds and specialized insect caterpillars synchronize their breeding cycles to match the precise emergence of young leaves and flower nectar. Phenological mismatches cause caterpillar food shortages and chick starvation.
- **Urban Biodiversity Lifecycle Support:** Knowing the fruiting calendar of native trees allows urban municipal corporations to plant complementary species, ensuring an uninterrupted year-round supply of nectar and fruits for urban birds and bats.
- **Fostering Scientific Observational Rigor:** Cultivates meticulous observational habits among Standard XI students, transforming passive walkers into active natural history researchers.

Phenology proves that the rhythmic pulses of trees govern the heartbeat of the entire terrestrial ecosystem.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

The phenological investigation was guided by structured botanical and observational objectives:

Structured Project Objectives :-

- • **Tree Selection & Geo-Tagging:** To identify, tag, and photograph 10 mature tree specimens representing different families in the local neighborhood.
- • **Seasonal Phenophase Monitoring:** To monitor and record six distinct phenophases (Leaf Emergence, Full Foliage, Flower Budding, Peak Blooming, Fruit Setting, and Leaf Senescence) at regular bi-weekly intervals.
- • **Associated Fauna Documentation:** To observe and document all avian, insect, and mammalian visitors utilizing each tree during specific flowering and fruiting phenophases.
- • **Native vs. Exotic Phenological Comparison:** To compare the seasonal resilience and wildlife support capacity of native species (e.g. Neem, Peepal, Babul) versus exotic species (e.g. Gulmohar, Subabul).
- • **Urban Arboriculture Guidelines:** To synthesize practical recommendations for urban municipal gardening based on seasonal biodiversity value.

The scope encompasses bi-weekly field logs, photographic phenophase archives, and taxonomic verification.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A standardized botanical phenology tracking protocol was deployed over consecutive seasons:

Methodological Protocol Executed :-

- **Phase 1 – Specimen Tagging & Baseline Recording:** Selected 10 healthy, mature tree specimens located along a fixed 800-meter avenue and campus transect; recorded height, girth at breast height (GBH), and GPS coordinates.
- **Phase 2 – Bi-Weekly Field Observations:** Visited all 10 trees every second Saturday morning between 7:00 AM and 9:00 AM to record the percentage of canopy undergoing specific phenophases.
- **Phase 3 – Fauna & Nesting Surveys:** Spent 15 minutes per tree recording insect pollinators (bees, butterflies), avian foragers, and active bird nest structures.
- **Phase 4 – Local Weather Data Correlation:** Recorded daily maximum and minimum temperatures and rainfall events from regional meteorological bulletins to correlate with budding events.
- **Phase 5 – Synthesis & Curricular Alignment:** Synthesized observational data into a comprehensive phenology matrix and correlated findings with Chapter 7 EVS competencies.

All floral and faunal species were authenticated using BNHS and Maharashtra State Flora references.

5. Observations (निरीक्षणे)

Bi-weekly field tracking across seasons yielded comprehensive phenological data for all ten monitored tree specimens:

5.1 Phenological Calendar & Wildlife Association Matrix

Tree Species	Leafing Period	Flowering Period	Fruiting Period	Observed Fauna & Nesting
Peepal (<i>Ficus religiosa</i>)	Copper-pink flush in March-April	Inconspicuous (internal syconia)	April-June; secondary crop in Nov	Flocks of Coppersmith Barbets, Parakeets, and Fruit Bats feeding on figs; Asian Koel calling.
Neem (<i>Azadirachta indica</i>)	Fresh pale green in March	March-April (fragrant white)	May-July (yellow drupes)	Honeybees and hoverflies pollinating; Bulbuls and Babblers dispersing seeds.
Gulmohar (<i>Delonix regia</i>) [Exotic]	Deciduous drop Jan-Feb; leaves May	April-June (scarlet flowers)	Persistent woody pods July-Dec	Few honeybees visiting blossoms; fragile branches with zero bird nests recorded.
Palas / Flame of Forest (<i>Butea monosperma</i>)	Leaves shed completely Jan-Feb	Feb-March (vermilion orange)	April-June (flat grey pods)	Purple Sunbirds, Mynas, and Drongos feeding on copious nectar; heavy bird congregation.
Karanj (<i>Pongamia pinnata</i>)	Bright green in April	April-May (lilac-white pea flowers)	Woody pods mature Jan-March	Larval host for Common Banded Awl butterfly; Tailorbird nest stitched on two leaves.
Rain Tree (<i>Samanea saman</i>) [Exotic]	Leaves fold at dusk & in rain	March-May & Sept-Nov (pink puffs)	Thick black pods Dec-Feb	Carpenter bees buzzing around flowers; common roosting tree for Black Kites.
Amaltas / Bahava	Leaves shed	April-June	Long	Emigrant butterflies laying eggs; Indian Palm

6. Analysis of data (माहितीचे विश्लेषण)

Detailed analysis of the phenological calendar reveals critical ecological synchronies between tree biology and urban fauna:

Core Analytical Inferences :-

- **Famine Food Reservoirs:** Native deciduous trees like Palas (*Butea monosperma*) and Semal (*Bombax ceiba*) bloom during the driest, hottest months (February-March) when water is critically scarce. Their large, cup-shaped flowers produce copious watery nectar that functions as an essential life-support drink for over 15 bird and insect species.
- **The Supremacy of Keystone Ficus Species:** Peepal and Banyan produce asynchronous fruiting crops multiple times a year. When other trees are barren in summer, mature Ficus trees carry thousands of ripe figs, sustaining entire urban bird populations.
- **Ecological Failure of Fast-Growing Exotics:** The study proved that while exotic trees (Subabul and Gulmohar) grow rapidly, they support significantly lower wildlife interactions. Gulmohar branches are soft and brittle, and zero bird nests were built on them, unlike the sturdy, dense branches of native Neem and Karanj.
- **Phenological Shifts as Early Warning Signs:** Comparing local blooming dates with historic district gazetteer records showed that Amaltas and Neem bloomed approximately 10 days earlier than historical records, likely driven by urban heat island effects.

The analysis proves that urban forestry must select native species that offer sequential, year-round ecological sustenance.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Tracking tree phenology reveals the exquisite, finely balanced choreography of nature. Native trees are indispensable biological anchors whose seasonal cycles sustain bird migrations, insect pollination, and urban wildlife survival. Preserving diverse native arboriculture is essential for urban climate adaptation.

Action Checklist for Urban Arboriculture :-

- ❑ Mandate municipal planting of native nectar and fruit trees (Palas, Semal, Jamun, Peepal).
- ❑ Prohibit the monoculture planting of exotic, storm-vulnerable trees like Subabul and Conocarpus.
- ❑ Maintain school phenology observation logs to contribute to national climate monitoring portals.
- ❑ Protect mature hollow trees that provide critical nesting cavities for parakeets and owls.
- ❑ Ensure adequate soil basins around avenue trees to support healthy root aeration and moisture uptake.

8. Bibliography / References (संदर्भ सूची)

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3. SeasonWatch India. *Citizen Science Tree Phenology Monitoring Manual*. NCBS, Bengaluru.
4. Maharashtra State Board. (2020). *Environmental Education and Water Security (Std. XI)*. Balbharati, Pune.